LEARNER GUIDE

Using and checking formulae in context

Formulae and Rearranging · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A formula always gives a number — but the number is only right if you chose the right formula, put the values in the right units, and worked accurately. This lesson uses six questions that solve almost any formula problem: **What do I know? What do I need? Substitute or rearrange? Are the units consistent? Calculate. Does the answer make sense — and does it fit when I put it back in?**

What you are learning to notice

- Check the units of every value before you substitute: a speed in km per hour needs a time in hours.
- Picture the answer in real life. A 21 m door or a 400 km walk tells you something has gone wrong.
- Substitute your answer back into the original formula to check it.
- Most silly answers come from a decimal point, mixed units, or multiplying instead of dividing.

Moving through it, one step at a time

Write down what you know and what you need. Decide whether to substitute straight in or rearrange first. Convert any values whose units do not match. Calculate, then check: is it sensible, and does it fit the original formula?

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Units first

A train travels 150 km in 90 minutes. Use s = d/t to find its speed in km per hour.

1. Know d = 150 km and t = 90 minutes; need s.	s is the subject: substitute
2. The answer is in km per hour, so the time must be in hours.	90 minutes = 1.5 hours
3. Substitute and calculate.	s = 150 ÷ 1.5 = 100
4. Check: sensible for a train? And 100 × 1.5 = 150 ✓	yes

100 km per hour.

Example 2 — Rearrange, then check

A taxi costs C = 3 + 2m pounds. How far can you go for £19?

1. Know C = 19; need m.	rearrange or substitute
2. Rearrange: m = (C – 3)/2.	m = (19 – 3)/2
3. Calculate.	m = 8
4. Check: 3 + 2 × 8 = 19 ✓	8 miles for £19 is sensible

8 miles.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (converting units of length, time and money, and Lessons 2 to 5 of this unit), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.